# Homework 07

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All work is in Python. Every random draw uses seed **432**.

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# Question 1: LDA

## Part (a): priors, class means, pooled covariance

With $K = 2$ classes the estimates are the class frequencies, the class centroids, and the within-class scatter divided by $n - K$: $\widehat{\pi}_k = n_k/n$, $\widehat{\boldsymbol{\mu}}_k = n_k^{-1} \sum_{y_i = k} \boldsymbol{x}_i$, and $\widehat{\boldsymbol{\Sigma}} = (n-K)^{-1} \sum_k \sum_{y_i = k} (\boldsymbol{x}_i - \widehat{\boldsymbol{\mu}}_k)(\boldsymbol{x}_i - \widehat{\boldsymbol{\mu}}_k)^{\mathsf{T}}$.

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```
n = 462, p = 9, K = 2;  n_0 = 302, n_1 = 160
priors: pi_0 = 0.6537, pi_1 = 0.3463
```

| | sbp | tobacco | ldl | adiposity | famhist | typea | obesity | alcohol | age |
|---|---|---|---|---|---|---|---|---|---|
| **mu_0 (chd=0)** | 135.460 | 2.630 | 4.340 | 23.970 | 1.320 | 52.370 | 25.740 | 15.930 | 38.850 |
| **mu_1 (chd=1)** | 143.740 | 5.520 | 5.490 | 28.120 | 1.600 | 54.490 | 26.620 | 19.150 | 50.290 |

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```
pooled covariance, 9 x 9, denominator n - K = 460
```

| | sbp | tobacco | ldl | adiposity | famhist | typea | obesity | alcohol | age |
|---|---|---|---|---|---|---|---|---|---|
| **sbp** | 405.400 | 14.600 | 4.600 | 49.200 | 0.300 | -15.600 | 18.900 | 64.400 | 95.100 |
| **tobacco** | 14.600 | 19.200 | 0.800 | 7.500 | 0.000 | -2.100 | 1.800 | 20.500 | 22.800 |
| **ldl** | 4.600 | 0.800 | 4.000 | 6.000 | 0.100 | 0.300 | 2.700 | -2.500 | 6.500 |
| **adiposity** | 49.200 | 7.500 | 6.000 | 56.800 | 0.400 | -5.300 | 22.700 | 16.100 | 60.500 |
| **famhist** | 0.300 | 0.000 | 0.100 | 0.400 | 0.200 | 0.100 | 0.200 | 0.800 | 1.000 |
| **typea** | -15.600 | -2.100 | 0.300 | -5.300 | 0.100 | 95.600 | 2.600 | 8.000 | -20.300 |
| **obesity** | 18.900 | 1.800 | 2.700 | 22.700 | 0.200 | 2.600 | 17.600 | 4.700 | 15.700 |
| **alcohol** | 64.400 | 20.500 | -2.500 | 16.100 | 0.800 | 8.000 | 4.700 | 598.300 | 27.900 |
| **age** | 95.100 | 22.800 | 6.500 | 60.500 | 1.000 | -20.300 | 15.700 | 27.900 | 184.100 |

The nine predictors sit on wildly different scales, so the diagonal runs from $0.23$ for the binary `famhist` to $598$ for `alcohol`. LDA needs no standardization because $\widehat{\boldsymbol{\Sigma}}^{-1}$ absorbs the scales.

## Part (b): in-sample prediction from hand-computed scores

Collect the linear part of the score in $\boldsymbol{W} = \widehat{\boldsymbol{\Sigma}}^{-1} \boldsymbol{M}^{\mathsf{T}}$ ($p \times K$, where $\boldsymbol{M}$ stacks the centroids) and the constants in $b_k = -\frac{1}{2} \widehat{\boldsymbol{\mu}}_k^{\mathsf{T}} \widehat{\boldsymbol{\Sigma}}^{-1} \widehat{\boldsymbol{\mu}}_k + \log \widehat{\pi}_k$. All scores are then the $n \times K$ matrix $\boldsymbol{X}\boldsymbol{W} + \boldsymbol{b}^{\mathsf{T}}$, and the fitted label is its row-wise maximiser.

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```
X (462, 9) @ W (9, 2) + b (2,)  ->  scores (462, 2)  ->  pred (462,) matches y (462,)
in-sample misclassification rate: 0.2532 = 117/462
```

| **predicted** | **0** | **1** |
|---|---|---|
| **observed** | | |
| **0** | 258 | 44 |

| predicted | 0 | 1 |
|---|---|---|
| **observed** | | |
| **1** | 73 | 87 |

The rule errs in one direction: of the $160$ true cases it recovers only $87$, because $\widehat{\pi}_1 = 0.346$ enters through $\log\widehat{\pi}_k$ and pushes the boundary toward the majority class. An in-sample error of $25.3\%$, against the $34.6\%$ a constant rule would achieve, is a fair warning that these nine predictors separate coronary heart disease only weakly.

## Part (c): comparison with `LinearDiscriminantAnalysis`

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```
sklearn priors: [0.6537 0.3463]   labels identical to part (b): True
largest |manual score difference - sklearn decision function|: 5.06e-14
```

| predicted | 0 | 1 |
|---|---|---|
| **observed** | | |
| **0** | 258 | 44 |
| **1** | 73 | 87 |

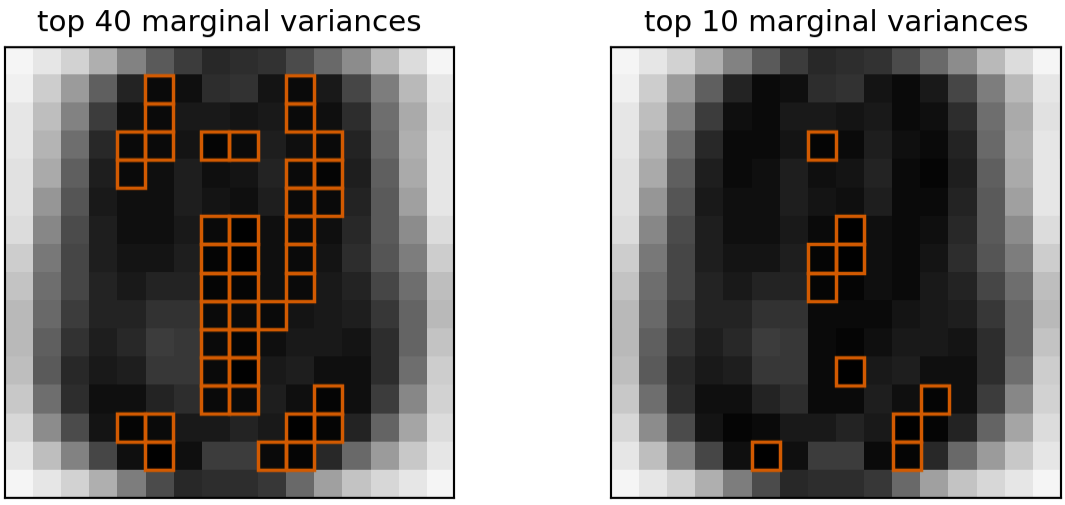
The results match. The confusion matrices are identical, and the agreement is not only at the level of labels: the hand-built score difference $s_1(\boldsymbol{x}) - s_0(\boldsymbol{x})$ equals `decision_function` to $5 \times 10^{-14}$, that is, to floating-point error. This is the expected outcome rather than a coincidence, because `solver="svd"` whitens the pooled within-class scatter using the same $n - K$ denominator as part (a) and leaves `priors=None` at the empirical frequencies, so it computes the same discriminant by a numerically safer route.

# Question 2: QDA and marginal screening

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```
train (7291, 256), test (2007, 256), classes [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
smallest digit class in training: n = 542
marginal variance: largest 0.800, 10th 0.761, 40th 0.721, smallest 0.00222
```

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Pixel-wise marginal standard deviation on the 16 x 16 grid (dark = more variable). Squares mark the screened pixels, 40 on the left and 10 on the right. Screening is unsupervised, so it concentrates on the central stroke region and keeps adjacent, nearly redundant pixels.

## Part (a): QDA with the 40 screened pixels

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```
test misclassification rate: 0.1555 = 312/2007    (training rate 0.0834)
```

| predicted | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|
| **observed** | | | | | | | | | | |
| **0** | 319 | 0 | 24 | 6 | 2 | 2 | 1 | 0 | 5 | 0 |
| **1** | 0 | 238 | 5 | 0 | 11 | 0 | 7 | 0 | 3 | 0 |
| **2** | 8 | 0 | 168 | 8 | 8 | 3 | 0 | 0 | 3 | 0 |
| **3** | 5 | 0 | 4 | 128 | 0 | 14 | 0 | 0 | 15 | 0 |
| **4** | 0 | 2 | 13 | 1 | 150 | 1 | 0 | 1 | 7 | 25 |
| **5** | 12 | 0 | 3 | 5 | 1 | 131 | 0 | 0 | 6 | 2 |
| **6** | 0 | 0 | 8 | 0 | 1 | 8 | 148 | 0 | 5 | 0 |
| **7** | 1 | 0 | 6 | 1 | 6 | 0 | 0 | 124 | 2 | 7 |
| **8** | 1 | 0 | 10 | 12 | 3 | 2 | 0 | 0 | 137 | 1 |
| **9** | 0 | 0 | 1 | 2 | 7 | 1 | 1 | 4 | 9 | 152 |

# Part (b): does the method work?

Yes. QDA fits without a singularity warning and reaches a $15.6\%$ test error, far better than the $90\%$ of a constant rule, so screening has turned an infeasible problem into a working one. The reason is a counting argument. QDA needs a separate invertible $m \times m$ covariance per class, which requires at least $m + 1$ observations in the smallest class and, for the inverse to be stable rather than merely defined, a good deal more. With all $m = 256$ pixels that fails outright: many pixels lie in the border region and are constant or nearly constant within a class — the smallest marginal variance over the whole training set is $0.002$ — so those within-class covariances are singular. With $m = 40$ the smallest digit class still has $542$ training images, roughly $13$ observations per dimension, and each of the ten $40 \times 40$ matrices is comfortably estimable. Screening on marginal variance also happens to discard exactly the offending predictors, since a pixel with no variance overall can have none within a class either.

The error is not spread evenly. The screened region covers the centre of the image, where strokes differ, and the confusion matrix shows the classical digit pairs: the worst classes are 4 ($25.0\%$, with $25$ of $200$ called 9) and 3 ( $22.9\%$, confused with 5 and 8), while 1 is nearly clean at $9.8\%$. The gap between $8.3\%$ training error and $15.6\%$ test error is the residual cost of fitting $10 \times \binom{41}{2} \approx 8\,200$ covariance parameters.

# Part (c): QDA with the 10 screened pixels

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```
test misclassification rate: 0.3423 = 687/2007    (training rate 0.2888)
40 predictors: test 0.1555, train 0.0834     10 predictors: test 0.3423, train 0.2888     ratio 2.20
```

| predicted | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|
| **observed** | | | | | | | | | | |
| **0** | 298 | 0 | 11 | 1 | 1 | 16 | 28 | 4 | 0 | 0 |
| **1** | 0 | 241 | 10 | 0 | 3 | 1 | 1 | 0 | 2 | 6 |
| **2** | 19 | 2 | 95 | 14 | 14 | 3 | 22 | 21 | 5 | 3 |
| **3** | 5 | 0 | 6 | 103 | 3 | 42 | 0 | 0 | 7 | 0 |
| **4** | 1 | 5 | 14 | 0 | 95 | 2 | 4 | 15 | 17 | 47 |
| **5** | 15 | 0 | 6 | 25 | 4 | 96 | 4 | 2 | 8 | 0 |
| **6** | 41 | 0 | 15 | 4 | 0 | 18 | 78 | 7 | 5 | 2 |
| **7** | 4 | 1 | 6 | 0 | 6 | 0 | 1 | 122 | 0 | 7 |
| **8** | 1 | 2 | 11 | 36 | 12 | 11 | 4 | 1 | 74 | 14 |
| **9** | 0 | 2 | 9 | 2 | 20 | 0 | 0 | 16 | 10 | 118 |

Test error more than doubles, from $15.6\%$ to $34.2\%$. This is not an estimation failure: ten dimensions are *easier* to estimate than forty, and the training error rises too, from $8.3\%$ to $28.9\%$. The model has stopped being able to represent the problem at all, which is bias rather than variance; with training and test error now within five points of each other, almost all of the error is already present in-sample.

Two features of the screening rule make $m = 10$ especially weak. It is unsupervised, ranking pixels by total variance with no reference to the digit label, so it selects pixels that vary a lot without checking that they vary *between* classes. And it ignores redundancy: four of the ten winners (pixels $104$, $119$, $120$, $135$) form a single touching cluster at the centre of the image and three more ($203$, $218$, $234$) form a second chain, with correlations of $0.68$ to $0.82$ between touching pairs, so the effective number of distinct measurements is well below ten. The confusion matrix shows the consequence as the collapse of specific pairs, with $41$ of $170$ sixes called 0 and $36$ of $166$ eights called 3. Marginal variance is a cheap filter that buys invertibility; it is not a good criterion for *which* predictors to keep once the budget is small.

# Question 3: Compare LDA and QDA

Class 1 has mean $(-1, 0)$ with correlation $+\rho$ between the two informative covariates, class 2 has mean $(1, 0)$ with correlation $-\rho$, and covariates 3 to 10 are independent standard normal noise in both classes. Both methods see all ten covariates with equal priors, and within each setting they are fitted to the identical training sample and scored on the identical test sample. Seed **432**.

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## Part (a): one dataset per $\rho$

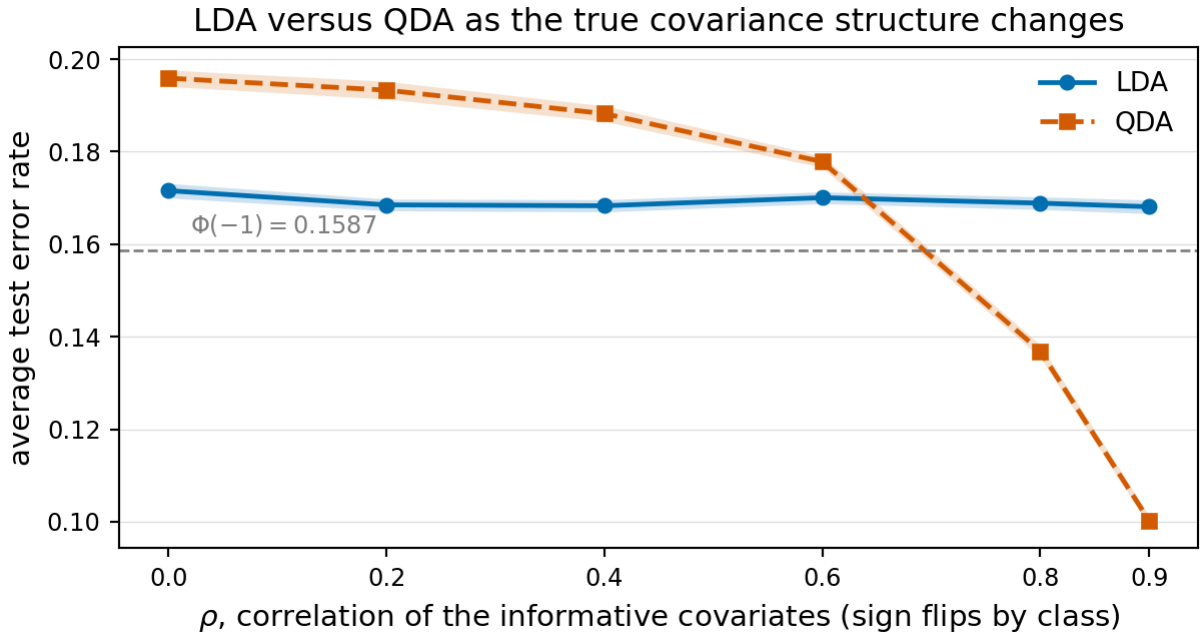
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| | LDA | QDA | difference |
|---|---|---|---|
| **rho** | | | |
| **0.000** | 0.171 | 0.201 | 0.030 |
| **0.200** | 0.172 | 0.181 | 0.009 |
| **0.400** | 0.158 | 0.172 | 0.014 |
| **0.600** | 0.153 | 0.161 | 0.007 |
| **0.800** | 0.181 | 0.149 | -0.032 |
| **0.900** | 0.181 | 0.103 | -0.079 |

A single replicate per setting already shows QDA overtaking LDA somewhere above $\rho = 0.6$, but the LDA column wanders between $0.154$ and $0.182$ with no trend. That is sampling noise in a 2000-point test set rather than an effect of $\rho$, and it is exactly why part (b) averages.

## Part (b): 50 replicates per $\rho$

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Average test misclassification rate over 50 replicates per setting, with 100 training and 1,000 test observations per class and 10 covariates. Bands are plus or minus one standard error of the mean. The dashed grey line is LDA's population error Phi(-1) = 0.1587.

| | LDA mean | QDA mean | QDA - LDA | QDA better in |
|---|---|---|---|---|
| **rho** | | | | |
| **0.000** | 0.172 | 0.196 | 0.024 | 0% of 50 runs |
| **0.200** | 0.169 | 0.193 | 0.025 | 0% of 50 runs |
| **0.400** | 0.168 | 0.188 | 0.020 | 0% of 50 runs |
| **0.600** | 0.170 | 0.178 | 0.008 | 18% of 50 runs |
| **0.800** | 0.169 | 0.137 | -0.032 | 100% of 50 runs |
| **0.900** | 0.168 | 0.100 | -0.068 | 100% of 50 runs |

## Part (c): why the curves cross

**LDA is flat and QDA improves.** LDA averages between $0.168$ and $0.172$ at every $\rho$, for a structural rather than numerical reason. Pooling averages the two class covariances, $\frac{1}{2}\begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} + \frac{1}{2}\begin{bmatrix} 1 & -\rho \\ -\rho & 1 \end{bmatrix} = \boldsymbol{I}$, so the $\rho$ terms cancel exactly. The pooled estimate converges to the identity whatever $\rho$ is, the fitted boundary is the hyperplane $x_1 = 0$ midway between the centroids, and the population error is $\Pr(X_1 > 0 \mid Y = 1) = \Phi(-1) = 0.1587$ for every $\rho$ — the dashed line in the figure. Pooling discards precisely the quantity that distinguishes the two classes here. QDA keeps it: the log-ratio of two normal densities with unequal covariances is quadratic, the true boundary is a hyperbola whose curvature grows with $\rho$, and the Bayes error falls, so QDA's average error drops from $0.196$ to $0.100$.

**At small $\rho$ the trade-off runs the other way.** At $\rho = 0$ the two class covariances really are identical, so LDA's restriction is correct and costs no bias, while QDA spends twice as many covariance parameters — $2 \times \binom{11}{2} = 110$ from $200$ training observations against LDA's $55$ — to estimate an equality it is never told about. That extra variance is the whole $2.4$-point gap ($0.196$ against $0.172$), and QDA loses in **all 50** replicates at $\rho \leq 0.4$. As $\rho$ grows LDA's error stays pinned at $\Phi(-1)$ while the Bayes error falls, so LDA's bias — its excess risk over the optimal rule — grows while QDA's variance cost stays fixed. The comparison tips once that bias exceeds the variance penalty: the curves cross near $\rho \approx 0.65$, and by $\rho \geq 0.8$ QDA wins in **all 50** replicates. Which method is better is a property of the data-generating distribution, not a ranking of the methods.

**The eight noise covariates cost both methods, and QDA about three times more.** They carry no signal, yet each adds parameters estimated from the same $200$ observations, and sample covariances between independent covariates are nonzero by chance, so the fitted boundary tilts into directions that are pure noise. Re-running the identical experiment with only the two informative covariates isolates the effect.

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| | LDA, p = 10 | LDA, p = 2 | LDA noise cost | QDA, p = 10 | QDA, p = 2 | QDA noise cost |
|---|---|---|---|---|---|---|
| **rho** | | | | | | |
| **0.000** | 0.172 | 0.161 | 0.010 | 0.196 | 0.162 | 0.034 |
| **0.200** | 0.169 | 0.160 | 0.008 | 0.193 | 0.161 | 0.033 |
| **0.400** | 0.168 | 0.161 | 0.008 | 0.188 | 0.162 | 0.026 |
| **0.600** | 0.170 | 0.159 | 0.011 | 0.178 | 0.151 | 0.026 |
| **0.800** | 0.169 | 0.158 | 0.011 | 0.137 | 0.118 | 0.018 |
| **0.900** | 0.168 | 0.159 | 0.009 | 0.100 | 0.086 | 0.015 |

At $\rho = 0$ dropping the junk improves LDA from $0.172$ to $0.161$, about one point, and QDA from $0.196$ to $0.162$, about $3.4$ points. The penalty is roughly three times larger for QDA because each junk covariate enters two covariance matrices rather than one, so its parameter count grows about twice as fast in $p$ and each estimate rests on half as much data. Irrelevant predictors are not free, and they are most expensive for the more flexible model.

# Question 4: Create and use an LDA/QDA analysis skill

## Part (a): defining the skill

The skill lives at `.claude/skills/stat432-lda-qda/SKILL.md` and is **4,052 bytes**, inside the $5,000$-byte limit; it is reproduced in full below. Its content is written entirely in terms of a supplied response and supplied predictors, with no dataset, variable name or expected result mentioned, so the same file applies unchanged to any classification problem. Three design choices carry most of the weight.

*The candidate list is closed.* LDA, QDA, PCA-reduced versions for $k = 2, \ldots, p-1$, and the blend $(1-a)p_{\text{LDA}} + a\,p_{\text{QDA}}$ over a grid of $a$ are allowed, and the skill says explicitly to add nothing else. The blend is the useful extension because its endpoints $a = 0$ and $a = 1$ *are* LDA and QDA, so a single grid interpolates between the pooled and per-class covariance assumptions and the two required comparisons come for free.

*The protocol is ordered so the test set cannot leak.* Split first; build one fold set and reuse it for every candidate; estimate centering, scaling and PCA loadings inside each training fold; select; then refit and touch the test set exactly once. The clause requiring identical folds across candidates is what makes the comparison a paired one.

*Selection uses the one-standard-error rule.* The skill requires the fold-to-fold standard error to be reported and the simplest candidate within one SE of the best to be chosen, counting complexity as $k(k+1)/2$ per covariance matrix. Without it the agent reports whichever grid cell won by noise, which is the specific failure the first draft produced.

## Parts (b) and (c): applying the skill, and the revisions it prompted

I invoked the skill explicitly, as a slash command naming it plus the three pieces of information it demands and nothing else:

> `/stat432-lda-qda` — Run the study on `homework/week-07/data/saheart.csv`. Response `chd`; predictors the nine other columns, with `famhist` coded 1 for Absent and 2 for Present. Seed 432. Seek the best predictive performance available within the methods you allow.

Explicit invocation is enforced through the skill's `description`, the only part of the file loaded into context before use. It states that the skill is for a dataset the student supplies and closes with *"Use this skill only when the student explicitly asks for it by name"* and *"Never start it on your own"*, so it is not triggered by ambient work on

classification; the alternative the question allows, keeping the file outside `.claude/skills/` and pasting it into the request, would serve equally well.

Checking the run produced two revisions, both justified by training and CV results alone. The first draft asked for plain 10-fold CV, and the grid it returned separated the top candidates by about $0.02$ percentage points against a fold standard error near one point — a fraction of a single observation, so the "winner" was noise. I changed the folds to five repeats of 10-fold and added the one-SE rule with an explicit parameter count. The second revision came from a genuine tie at $20$ covariance parameters, so the skill now breaks ties by lower CV error.

## The complete `SKILL.md`

File size **4,052 bytes**.

```
---
name: stat432-lda-qda
description: Run a discriminant-analysis study (LDA, QDA, PCA-reduced versions, and LDA/QDA probability blends) on a dataset the student
         supplies and write a one-page report. Use this skill only when the student explicitly asks for it by name and supplies the data
         file, the response, and the predictors. Never start it on your own.
---

# Discriminant Analysis Study

The request supplies the data file, the response column, the predictor columns, and any
coding. Ask for whatever is missing instead of guessing. Everything below is fixed and does
not depend on the dataset.

## Candidate classifiers

Use only these, each with empirical class priors unless the request says otherwise:

- **LDA** on the supplied predictors: one pooled covariance, denominator $n-K$.
- **QDA** on the supplied predictors: one covariance per class.
- **PCA + LDA** and **PCA + QDA** on the leading $k$ principal components, $k = 2,\dots,p-1$.
- **Blend** of the two class-probability vectors, $(1-a)\,p_{\text{LDA}} + a\,p_{\text{QDA}}$,
  on the same inputs, with $a$ on a grid from 0 to 1 in steps of 0.25. The ends $a=0$ and
  $a=1$ reproduce LDA and QDA, so always report those two as the reference points.

Add nothing else: no logistic regression, trees, boosting, or penalized fits.

## Protocol

1. **Split once.** Hold out a stratified test fraction (default 30%) before looking at any
   candidate. Report the seed. Touch the test set once, at step 5.
2. **One fold set.** Build stratified folds on the training data (default 10 splits, 5
   repeats) and reuse the identical folds for every candidate, so that differences between
   candidates are not fold noise.
3. **Preprocess inside the fold.** Estimate centering, scaling, and PCA loadings from the
   training part of each fold only, then apply those constants to the held-out part. Never fit
   preprocessing on the full training set before cross-validation.
4. **Select.** Score candidates by cross-validated misclassification rate and report the
   fold-to-fold standard error. Apply the one-standard-error rule: among candidates within one
   SE of the lowest CV error, choose the one estimating the fewest covariance parameters,
   counting $k(k+1)/2$ per covariance matrix and breaking ties by lower CV error. Also name the
   minimum-CV candidate when it differs.
5. **Refit and score once.** Refit the selected preprocessing and classifier on all training
   observations, predict the held-out test set a single time, and report the test
   misclassification rate. Revise nothing after this step.

Never select, tune, or revise a choice using test-set results.

## Report

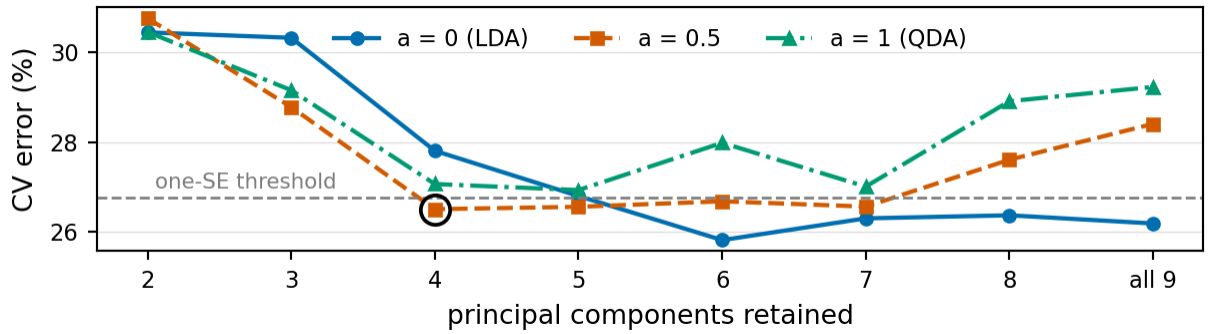
At most one printed page, in this order, keeping every item.

- **Setup**, 2-3 sentences: $n$, $p$, class sizes, the split, the seed, the fold scheme.
- **One figure**: CV error against model complexity, with one curve per blend weight and the
  one-SE threshold as a horizontal reference line. Label both axes with units, use a
  colorblind-safe palette, and give each curve its own line type so the figure survives
  grayscale.
- **One table**: one row per reported candidate, with CV error, its standard error, the
  covariance-parameter count, and a test error only for models actually scored on the test
  set. Mark the selected row.
- **Numerical results**: the selected candidate, its CV error with SE, its test error, and the
  test confusion matrix.
- **Findings**, 3-5 sentences: which covariance assumption the data favor, pooled or
  per-class, and what the comparison says about bias and variance at this $n$ and $p$.
- **Limitations**, 2-3 sentences: give the test-set standard error
  $\sqrt{\hat e(1-\hat e)/n_{\text{test}}}$ and say whether the reported gaps exceed it. Name
  the assumptions doing the work, within-class normality and the covariance structure, and any
  class imbalance.

Report error rates to the nearest 0.1 percentage point. If any covariance is singular or a fit
warns about collinearity, say which candidate and report it; never drop a candidate silently.
```

# LDA/QDA analysis of the `SAheart` data

**Setup.** The response is `chd` on $n = 462$ observations with $p = 9$ numerical predictors (`famhist` coded 1/2), split $302$ without and $160$ with coronary heart disease. A stratified $70/30$ split at seed **432** reserves $139$ test observations $(91/48)$ and leaves $323$ for training $(211/112)$. All candidates share one set of $5 \times 10$-fold stratified folds $(50$ fits each), with standardization and PCA loadings estimated inside each training fold only.




CV error against retained principal components, one curve per blend weight a (a = 0 is LDA, a = 1 is QDA). Rightmost points use all nine predictors untransformed. Grey line is the one-SE threshold; the open circle is the selected candidate.

| candidate | CV error (%) | SE (%) | cov. params | test error (%) |
|---|---|---|---|---|
| **LDA, all 9 predictors** | 26.2 | 0.9 | 45 | 26.6 |
| **QDA, all 9 predictors** | 29.2 | 0.9 | 90 | 28.1 |
| **blend a = 0.5, all 9 predictors** | 28.4 | 0.9 | 90 | |
| **PCA k = 2 + LDA** | 30.4 | 1.1 | 3 | |
| **minimum CV: PCA k = 6 + LDA** | 25.8 | 1.0 | 21 | 26.6 |
| **> selected: PCA k = 4 + blend a = 0.5** | 26.5 | 1.0 | 20 | 24.5 |

**Numerical results.** The grid of $8$ dimension settings $\times$ $5$ blend weights has its minimum at PCA $k = 6$ with $a = 0$ (pure LDA), $25.8\%$ CV error with a fold SE of $1.0$ point, so the one-SE threshold is $26.8\%$. Sixteen of the forty candidates fall below it, so the one-SE rule governs the choice and takes the cheapest: **PCA** $k = 4$ **with blend** $a = 0.5$, $20$ covariance parameters, CV error $26.5\%$ $(\pm 1.0)$. Refitted on all $323$ training observations and scored once on the reserved test set it misclassifies $24.5\%$ $(34/139)$, against $26.6\%$ for ordinary LDA and $28.1\%$ for ordinary QDA, with test confusion matrix

```
predicted   0   1
observed
0          79  12
1          22  26

sensitivity 0.542   specificity 0.868   test SE 3.6 points
```

**Findings.** The data favour the pooled covariance. Ordinary QDA is the worst candidate in the grid at $29.2\%$ CV error, $3.0$ points above ordinary LDA and more than three fold SEs away — the one difference here that noise does not explain: at $n_{\text{train}} = 323$ with nine correlated predictors, QDA's two $9 \times 9$ covariances cost $90$ parameters against LDA's $45$ and the extra flexibility buys no bias reduction. Where the blend helps at all it is at reduced dimension, once PCA has cut the per-class covariances to $k = 4$ and made the quadratic term cheap enough to estimate. Beyond that the comparison is flat, everything from $k = 4$ upward sitting within about one point, so dimension reduction mainly removes parameters without removing signal. Sensitivity $0.542$ against specificity $0.868$ is the familiar consequence of a $0.5$ cutoff under a $35\%$ positive rate.

**Limitations.** With $n_{\text{test}} = 139$ the test SE is $3.6$ points, so the $2.1$-point margin of the selected model over ordinary LDA is inside noise and is not a demonstrated gain; only the LDA-over-QDA gap is supported, and by CV rather than by the test set. Both methods assume within-class normality, which `tobacco` and `alcohol` violate badly as both are right-skewed with a spike at zero, so the fitted probabilities are better read as scores than as calibrated risks. A single split also leaves one split's worth of variability in the test estimate, and $0.5$ is an arbitrary cutoff at a $35\%$ positive rate.